

the algae or cyanobacteria by protecting them from the environment by its filaments, which also gather moisture and nutrients from the environment, and (usually) provide an anchor to it.

192. In lichen, a chlorophyll-containing partner, ____ live together.

- (a) alga and a fungus
- (b) fungus and a bacterium
- (c) alga and a virus
- (d) More than one of the above
- (e) None of the above

68th B.P.S.C. (Pre) 2022

Ans. (a)

See the explanation of above question.

193. Which of the following do not grow in polluted area?

- (a) Pseudomonas
- (b) Algae
- (c) Lichen
- (d) Gymnosperms

Rajasthan P.C.S. (Pre) 2023

Ans. (c)

Lichens usually grow in a wide range of environmental conditions. But they are quite sensitive to atmospheric pollutants like sulphur dioxide and nitrogen. To grow, lichens get their nutrients from the air. Because lichens have no roots or protective surface, they cannot filter what they absorb, so anything in the air is taken straight inside. If there are pollutants, it can accumulate in the lichen and can become toxic very quickly. Air pollution also leads to a decline in the lichen population and thus, they can be used as indicator organisms. As a rule of thumb, the smaller the size and less variety of lichens in the area, the more polluted is the environment.

194. Nitrogen fixing bacteria present in the nodules of the root of leguminous plant are–

- (a) Saprophytic
- (b) Parasitic
- (c) Symbiotic
- (d) Protolytic

Uttarakhand P.C.S. (Pre) 2007

Ans. (c)

Nitrogen fixing bacteria present in the nodules on the roots of the leguminous plant are symbiotic. During nitrogen fixation, bacteria such as *Rhizobium japonicum* convert nitrogen gas (N_2) into ammonium compounds (NH_4OH). These bacteria live in the roots of legumes including alfalfa, clover, peas and beans. Some blue-green algae can also fix nitrogen.

Cell

Notes

- The cell (from Latin-**Cella**- meaning small room) is the basic structural, functional and biological unit of all known living organisms.
 - A cell is the smallest unit of life. Cells are often called the ‘**building blocks**’ of life.
 - The study of cells is called **cell biology** or **cellular biology**.
 - The cell consists cytoplasm enclosed within a membrane, which contains many biomolecules such as **proteins** and **nucleic acid**.
 - The number of cells in plants and animals varies from species to species. Human contain around 40 trillion (4×10^{13}) cells.
 - Most plant and animal cells are visible only under a microscope, with dimensions between 1 to 100 micrometres.
 - Cells were discovered by **Robert Hooke** in 1665 (detailed description in a book written by him – **Micrographia**).
 - Cell theory, first developed in 1839, by **Schleiden** and **Schwann**. According to his theory, all organisms are made of one or more cell and all cells originate from pre-existing cells.
 - Cells emerged on Earth at least 3.5 billion years ago.
 - On the basis of absence or presence of a **nucleus**, the cells have been divided into two groups-
- I . Prokaryotic cells :**
- **Prokaryotic** cells are cells that do not have a true nucleus or most cell organelles. Organisms that have prokaryotic cells are unicellular and called **prokaryotes**.
 - Prokaryotic cells do not have a true nucleus that contains their genetic material as eukaryotic cells do. Instead, prokaryotic cells have a nucleoid region, which is an irregularly shaped region that contains the DNA and is not surrounded by the membrane. The cell wall is present in prokaryotes.
 - The ribosome is found in cytoplasm.
 - Histone protein is not annexed with the nucleic acid.
 - Such type of cells is found in bacteria and blue green algae. Prokaryotes include bacteria and archaea, two of the three domains of life. There cells were the first form of life on Earth.
- II . Eukaryotic cells :**
- Eukaryotic cells are cells that contain a nucleus and organelles, which are enclosed by a plasma membrane.

- Organisms that have eukaryotic cells include protozoa, algae, slime moulds, fungi, plants and animals. These organisms are grouped into the biological domain **Eukaryota**.
 - Eukaryotic cells are large and more complex than prokaryotic cells. These cells are about fifteen times wider than a typical prokaryote and can be as much as a thousand times greater in volume.
 - Eukaryotic cells contain a variety of structures called organelles, which perform various functions within the cell which are as follows –
- (a) Cell wall :**
- Cell wall is found in the cells of plants and algae, fungi and most chromalveolates, which is a layer outside the cell membrane, providing the cell with structural support, protection, and a filtering mechanism.
 - Plants cell wall are made up of cellulose, hemicellulose and pectin. The fungi cell wall is made up of **chitin**.
 - Animal cells lack a cell wall.
- (b) Plasma membrane/Cell membrane :**
- The plasma membrane is a biological membrane that separates the interior of all cells from the outside environment, which protects the cell from its environment, consisting of lipid bilayer with embedded protein.
 - The cell membrane is a **semipermeable** or **selectively permeable membrane** which controls the movements of ions and organic molecules.
- (c) Endoplasmic Reticulum (ER) :**
- ER is a cellular organelle. It is the transport network for molecules going to specific places.
 - ER is of two types –
- (i) Rough Endoplasmic Reticulum (RER) –** The ribosomes are found on its surface (which gives it a rough appearance) so it acts as a site for protein synthesis.
- (ii) Smooth Endoplasmic Reticulum (SER) –** It is without ribosome. It synthesizes lipids, phospholipids and steroids. Similar to the SER, **Sarcoplasmic Reticulum (SR)** is found only in muscle cells (myocytes). The SR stores and pumps calcium ions.
- (d) Ribosome :**
- The ribosome is a cell organelle. Ribosomes are found free in the cytoplasm or bound to the endoplasmic reticulum to form rough ER. It functions as a micromachine for making proteins.
- (e) Mitochondria :**
- Mitochondria are surrounded by two membranes and have their own genome.
 - Mitochondria acts as the power plants, it means that the maximum energy in form of ATP is produced within the mitochondria, hence it is also known as **Power House** of the cells.
 - During cellular respiration, the initial process takes place in cytoplasm and then in mitochondria.
 - The most prominent roles of mitochondria are to produce the energy currency of the cell, **ATP** (adenosine triphosphate), through respiration, and to regulate cellular metabolism. During cellular respiration most ATP is formed in Krebs's cycle (Citric acid cycle).
- (f) Golgi body :**
- Golgi body is also known as **Golgi complex** or **Golgi apparatus**. It was identified in 1897 by **Camillo Golgi**.
 - The Golgi complex is responsible for manufacturing, warehousing and shipping certain cellular products such as proteins and lipids that are synthesized by the cell. Its functional unit is called **cisternae**.
- (g) Lysosomes :**
- Lysosomes are also known as the **suicidal bag** of cells.
 - These are membrane-enclosed organelles that contain an array of enzymes capable of breaking down all types of biological polymers-proteins, nucleic acids, carbohydrates and lipids. It functions on the digestive system of the cell serving both to degrade material taken up from outside the cell and to digest absolute components of the cell itself.
 - Lysosomes also take part in killing the viruses and bacteria which have entered within the cell.
 - The hydrolyzing enzymes act in acidic medium.
 - After the rupturing of membrane of lysosome, the enzyme came out, which hydrolyzes all components of cell wall by the process hydrolysis.
- (h) Centrosome :**
- It is the cytoskeleton organiser of the cell. It produces the microtubules of a cell – a key component of the cytoskeleton.
 - A single centrosome is present in the animal cells. Plants and most fungi lacks centrosome.
- (i) Vacuoles**
- Vacuoles sequester waste products and in plant cells store water.
 - The vacuoles of plant cells and fungal cells are usually larger than those of animal cells.
- Note :** By scientific convention, the term **lysosome** is applied to these vesicular organelles only in animals and the term **vacuole** is applied to those in plant, fungi and algae (some animal cells also have vacuoles). However, some botanists argued that these vacuoles are lysosomes.

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(j) Nucleus :

- Nucleus was discovered by **Robert Brown** in 1831.
- The nucleus is a membrane-bound organelle found in eukaryotic cells. Eukaryotes usually have a single nucleus

but a few cell types, such as mammalian R.B.C. have no nuclei and few others as **Vaucheria** and **Cladophora** has many nuclei.

- It is the controlling and coordinating organ of the nucleus due to the presence of DNA in it. It houses the cell's chromosomes, and is the place where almost all DNA replication and RNA synthesis occur.
- The nucleus is spherical and separated from the cytoplasm by a double membrane called the nuclear envelope.
- DNA is also found in mitochondria and chloroplast (found only in plant cells).

Programmed Cell Death (PCD) :

- Programmed cell death is the death of a cell in any form, mediated by an intracellular program, and is also referred to as **cellular suicide**.
- PCD is carried out in a biological process, which usually confers advantage during an organism's life cycle. For example, the differentiation of fingers and toes in a developing human embryo occurs because cell between the fingers apoptose ; the result is that the digits are separate.
- PCD serves fundamental functions during both plant and animal tissue development.
- **Apoptosis** and **autophagy**, both are the forms of programmed cell death.

Question Bank

1. What is the structural and functional unit of life?

- | | |
|------------|--------------|
| (a) Cell | (b) Organ |
| (c) Tissue | (d) Organism |

70th B.P.S.C. (Pre) 2024

Ans. (a)

The structural and functional unit of life is the cell, which is the fundamental building block of all living organisms, responsible for carrying out all life processes. All living organisms, from the simplest bacteria to complex animals, are composed of cells. Cells are responsible for both the structure and function of an organism. They provide the building blocks for tissues, organs, and organ systems, and they carry out all the processes necessary for life, such as respiration, nutrient uptake, and waste removal. Cells are organized into tissues, which are then organized into organs, and finally into organ systems, which work together to form a complete organism.

2. Which organelle in the cell, other than the nucleus, contains DNA ?

- | | |
|---------------|---------------------|
| (a) Centriole | (b) Golgi apparatus |
| (c) Lysosome | (d) Mitochondrion |

I.A.S. (Pre) 2001

Ans. (d)

Besides the nucleus, DNA is also found in the mitochondria and chloroplast of cells, which is a genetic material and carries the symptom signs of creatures to descendants from a parent like a horoscope.

3. Which one of the following statements is correct?

- (a) Prions are the smallest free-living cells.
- (b) The cell wall of Mycoplasmas is made up of amino sugars.
- (c) Viroids consist of single-stranded RNA molecules
- (d) Rickettsias lack cell wall

I.A.S. (Pre) 2002

Ans. (c)

Prions are smallest proteinaceous infectious particles. Mycoplasma is a genus of bacteria that lack a cell wall around their cell membrane. Viroids are the smallest infectious pathogens known consisting solely of short strands of circular, single-stranded RNA without protein coats. They are mostly plant pathogens, some of which are of economic importance. Rickettsia is a genus of non-motile, gram-negative, non-spore forming, highly pleomorphic bacteria. They have cell wall.

4. Which of the following statements are correct regarding the general difference between plant and animals cells?

- 1. Plant cells have cellulose cell walls whilst animal cells do not.
- 2. Plant cells do not have plasma membrane unlike animal cells which do.
- 3. Mature plant cell has one large vacuole whilst animal cell has many small vacuoles.

Select the correct answer using the code given below:

- (a) 1 and 2 only
- (b) 2 and 3 only
- (c) 1 and 3 only
- (d) 1, 2 and 3

I.A.S. (Pre) 2020

Ans. (c)

Cell wall is found in the cells of plants, which is made up of cellulose. It provides the cell structural support, protection and a filtering mechanism. Animal cells lack cell wall. Thus, statement 1 is correct. The plasma membrane is a biological membrane that separates the interior of all cells from the outside environment which protects the cell from its environment. Animal cells as well as plant cells have plasma membrane. Hence, statement 2 is incorrect. A vacuole is a membrane - bound cell organelle which are enclosed compartments filled with water containing organic and inorganic molecules. Most mature plant cells have one large vacuole while an animal cell contains several small vacuoles performing various functions. Therefore, statement 3 is correct.

5. How does the plant cell differ from the animal cell?

- (a) Mitochondria
- (b) Cell wall
- (c) Presence of protoplasm
- (d) Cell nucleus

U.P.P.C.S. (Pre) 1991

Ans. (b)

A plant cell is different from animal cell mainly on the basis of the cell wall. The outer shell in a plant cell (which is made from cellulose in green plants) is known as cell wall, which is not found in the animal cell. Except this, green plastids are also found in plant cells while not in animal cells.

6. What is not found in the animal cell?

- (a) Cell wall of cellulose
- (b) Nucleus
- (c) Mitochondria
- (d) None of these

M.P.P.C.S. (Pre) 2010

Ans. (a)

See the explanation of above question.

7. Which part of the animal cell is called power house?

- (a) Cell wall
- (b) Nucleus
- (c) Mitochondria
- (d) Entire cell

M.P.P.C.S. (Pre) 2010

Ans. (c)

Animal cells and plant cells both contain cell membrane, cytoplasm, and nucleus. The nucleus has two major functions. It stores the cell's hereditary material or DNA and it co-ordinates the cell's activities within cells. The mitochondrion (plural mitochondria) is a membrane-bound organelle found in most eukaryotic cells. Mitochondria range from 0.5 to 1.0 μ in diameter. These structures are sometimes described as 'the powerhouse of the cell' as they generate energy through respiration. During cellular respiration, the initial process takes place in cytoplasm and then in mitochondria.

8. Which of the following organelle is called 'Power House' of the cell?

- (a) Ribosome
- (b) Mitochondria
- (c) Cell membrane
- (d) Centrosome

U.P.R.O./A.R.O. (Mains) 2013

Ans. (b)

See the explanation of above question.

9. The powerhouse of cell is :

- (a) Cell wall
- (b) Mitochondria
- (c) Ribosome
- (d) Nucleus
- (e) None of the above/More than one of the above

64th B.P.S.C. (Pre) 2018

Ans. (b)

See the explanation of above question.